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Equity Valuation: Concepts and Basic Tools

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MULTIPLER MODELS AND RELATIONSHIP AMONG PRICE MULTIPLES, PRESENT VALUE MODELS, AND FUNDAMENTALS

Learning Outcomes

- explain the rationale for using price multiples to value equity, how the price to earnings multiple relates to fundamentals, and the use of multiples based on comparables
- calculate and interpret the following multiples: price to earnings, price to an estimate of operating cash flow, price to sales, and price to book value
- explain advantages and disadvantages of each category of valuation model

The term **price multiple** refers to a ratio that compares the share price with some sort of monetary flow or value to allow evaluation of the relative worth of a company's stock. Some practitioners use price ratios as a screening mechanism. If the ratio falls below a specified value, the shares are identified as candidates for purchase, and if the ratio exceeds a specified value, the shares are identified as candidates for sale. Many practitioners use ratios when examining a group or sector of stocks and consider the shares for which the ratio is relatively low to be attractively valued securities.

Price multiples that are used by security analysts include the following:

- Price-to-earnings ratio (P/E). This measure is the ratio of the stock price to earnings per share. P/E is arguably the price multiple most frequently cited by the media and used by analysts and investors (**Block 1999**). The seminal works of **McWilliams (1966)**, **Miller and Widmann (1966)**, **Nicholson (1968)**, **Dreman (1977)**, and **Basu (1977)** presented evidence of a return advantage to low-P/E stocks.
- Price-to-book ratio (P/B). The ratio of the stock price to book value per share. Considerable evidence suggests that P/B multiples are inversely related to future rates of return (**Fama and French 1995**).
- Price-to-sales ratio (P/S). This measure is the ratio of stock price to sales per share. **O'Shaughnessy (2005)** provided evidence that a low P/S multiple is the most useful multiple for predicting future returns.
- Price-to-cash-flow ratio (P/CF). This measure is the ratio of stock price to some per-share measure of cash flow. The measures of cash flow include free cash flow (FCF) and operating cash flow (OCF).

A common criticism of all of these multiples is that they do not consider the future. This criticism is true if the multiple is calculated from trailing or current values of the divisor. Practitioners seek to counter this criticism by a variety of techniques, including forecasting fundamental values (the divisors) one or more years into the future. The resulting forward (leading or prospective) price multiples may differ markedly from the trailing price multiples. In the absence of an explicit forecast of fundamental values, the analyst is making an implicit forecast of the future when implementing such models. The choice of price multiple—trailing or forward—should be used consistently for companies being compared.

Besides the traditional price multiples used in valuation, just presented, analysts need to know how to calculate and interpret other ratios. Such ratios include those used to analyze business performance and financial condition based on data reported in financial statements. In addition, many industries have specialized measures of business performance that analysts covering those industries should be familiar with. In analyzing cable television companies, for example, the ratio of total market value of the company to the total number of subscribers is commonly used. Another common measure is revenue

valued securities.

Relationships among Price Multiples, Present Value Models, and Fundamentals

Price multiples are frequently used independently of present value models. One price multiple valuation approach, the method of comparables, does not involve cash flow forecasts or discounting to present value. A price multiple is often related to fundamentals through a discounted cash flow model, however, such as the Gordon growth model. Understanding such connections can deepen the analyst's appreciation of the factors that affect the value of a multiple and often can help explain reasons for differences in multiples that do not involve mispricing. The expressions that are developed can be interpreted as the *justified value* of a multiple—that is, the value justified by (based on) fundamentals or a set of cash flow predictions. These expressions are an alternative way of presenting intrinsic-value estimates.

As an example, using the Gordon growth model identified previously in Equation 9 and assuming that price equals intrinsic value ($P_0 = V_0$), we can restate Equation 9 as follows:

$$P_0 = \frac{D_1}{r - g} \qquad 14$$

To arrive at the model for the justified forward P/E given in Equation 15, we divide both sides of Equation 14 by a forecast for next year's earnings, E_1. In Equation 15, the dividend payout ratio, p, is the ratio of dividends to earnings:

$$\frac{P_0}{E_1} = \frac{D_1/E_1}{r - g} = \frac{p}{r - g} \qquad 15$$

Equation 15 indicates that the P/E is inversely related to the required rate of return and positively related to the growth rate; that is, as the required rate of return increases, the P/E declines, and as the growth rate increases, the P/E increases. The P/E and the payout ratio appear to be positively related. This relationship may not be true, however, because a higher payout ratio may imply a slower growth rate as a result of the company retaining a lower proportion of earnings for reinvestment. This phenomenon is referred to as the dividend displacement of earnings.

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EXAMPLE 11

Value Estimate Based on Fundamentals

1. Petroleo Brasileiro SA, commonly known as Petrobras, was once labeled "the most expensive oil company" by Bloomberg.com. Data for Petrobras and the oil industry, including the trailing twelve-month (TTM) P/E and payout ratios, appear below.

	Petrobras	Industry
P/E ratio (TTM)	39.61	13.0
Return on assets (TTM) (%)	3. 0	3.2
EPS 3-year growth rate (%)	NM	66.00
EPS (MRQ) vs. Qtr. 1 yr. ago (% change)	138.96	−12.0

Note: NM stands for non-quantifiable. Petrobras EPS has decreased from a loss of BRL 1.14 per share to a profit of BRL 0.16 per share in the most recent period. MRQ stands for "most recent quarter."
Source: Reuters.

Explain how the information shown supports a higher P/E for Petrobras than for the industry.

Hide Solution

Solution:

EPS for the period 2014 to 2017, and has paid no dividend during that period.

A higher payout ratio supports a higher P/E. Furthermore, to the extent that higher EPS growth implies a high growth rate in dividends, the high EPS growth rate supports a high P/E. The higher P/E ratio is due to an improvement in the underlying financial performance of the company and the expected higher growth potential of the stock compared to the median firms in the industry.

EXAMPLE 12

Determining Justified Forward P/E

Heinrich Gladisch, CFA, is estimating the justified forward P/E for Nestlé, one of the world's leading nutrition and health companies. Gladisch notes that sales for 2017 were SFr89.78 billion (US$90.3 billion) and that net income was SFr7.18 billion (US$7.25 billion). He organizes the data for EPS, dividends per share, and the dividend payout ratio for the years 2013–2017 in the following table:

	2013	2014	2015	2016	2017
Earnings per share	SFr3.24	SFr4.54	SFr2.90	SFr2.76	SFr2.32
Year over year % change		44.6%	–36.1%	–4.8%	–15.9%
Dividend per share	SFr2.15	SFr2.2	SFr2.25	SFr2.3	SFr2.35
Year over year % change		2.3%	2.3%	2.2%	2.2%
Dividend payout ratio	68.5%	48.5%	77.6%	83.3%	99.2%

Gladisch calculates that ROE averaged 15.5 percent in the period 2013−2017 but was below that level at 11.7 percent in 2017. In that year, however, Nestlé's reported net income included a large nonrecurring component. The company reported 2017 "underlying earnings," which it defined as net income "from continuing operations before impairments, restructuring costs, results on disposals and significant one-off items," to be SFr2.93, giving an adjusted 14.8% ROE. Predicting increasing improvement in Nestlé's profit margins from growth in its product markets, Gladisch estimates a long-run ROE of 21.5 percent.

Gladisch decides that the dividend payout ratios of the 2013–2016 period—averaging 67.7 percent—are more representative of Nestlé's future payout ratio than is the high 2017 dividend payout ratio (when based on reported earnings). The dividend payout ratio in 2017 was higher because management apparently based the 2017 dividend on the components of net income that were expected to continue into the future. But basing a dividend on net income including non-recurring items creates the potential need to increase dividends in the future. Rounding up the 2013−2017 average, Gladisch settles on an estimate of 68 percent for the dividend payout ratio for use in calculating a justified forward P/E using Equation 15.

Gladisch's firm estimates that the required rate of return for Nestlé's shares is 9 percent per year. Gladisch also finds the following data at the opposite ends of the spectrum of external research analyst forecasts:

	2018E	2019E
Most optimistic analyst forecast:		
EPS	SFr3.99	SFr4.33
Year over year % change	71.9%	8.5%
P/E (based on a target price of SFr105)	26.3	24.2
Least optimistic analyst forecast:		

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P/E (based on a target price of SFr68)	19.3	18.9

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1. Based only on information and estimates developed by Gladisch and his firm, estimate Nestlé's justified forward P/E.

Hide Solution

Solution to 1:

The estimate of the justified forward P/E is 32.38. The dividend growth rate can be estimated by using Equation 10 as (1 – Dividend payout ratio) × ROE = (1 – 0.68) × 0.21.5 = 0.069, or 6.9 percent. Therefore,

$$\frac{P_0}{E_1} = \frac{\text{Payout}}{r - g} = \frac{0.68}{0.09 - 0.069} = 32.38$$

2. Compare and contrast the justified forward P/E estimate from Question 1 to the estimates from each end of the spectrum of external research analysts forecasts.

Hide Solution

Solution to 2:

The estimated justified forward P/E of 32.38 is higher than the justified 2018 P/E estimates of 26.3 and 19.3 of the two analysts. Using a required rate of return of 9.5 percent rather than 9 percent results in a justified forward P/E estimate of 26.2 = 0.68/(0.095 – 0.069). Using an ROE of 16.5 percent (the average ROE of the 2013–2016 period) rather than 21.5 percent results in a justified forward P/E estimate of 18.4 = 0.68/[0.09 – (0.32)(0.165)] = 0.68/(0.09 – 0.053). The justified forward P/E is very sensitive to changes in the inputs.

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Justified forward P/E estimates can be sensitive to small changes in assumptions. Therefore, analysts can benefit from carrying out a sensitivity analysis, as shown in Exhibit 6, which is based on Example 12. Exhibit 6 shows how the justified forward P/E varies with changes in the estimates for the dividend payout ratio (columns) and return on equity. The dividend growth rate (rows) changes because of changes in the retention rate (1 – Payout rate) and ROE. Recall *g* = ROE times retention rate.

Exhibit 6:

Estimates for Nestlé's Justified Forward P/E (Required Rate of Return = 9 Percent)

	Dividend Payout Ratio				
Constant Dividend Growth Rate (%)	**55%**	**60%**	**65%**	**70%**	**75%**
4.0	11.0	12.0	13.0	14.0	15.0
4.5	12.2	13.3	14.4	15.6	16.7
5.0	13.8	15.0	16.3	17.5	18.8
5.5	15.7	17.1	18.6	20.0	21.4
6.0	18.3	20.0	21.7	23.3	25.0
6.5	22.0	24.0	26.0	28.0	30.0
7.0	27.5	30.0	32.5	35.0	37.5
7.5	36.7	40.0	43.3	46.7	50.0

MO Fix the spelling of the title: "Multipler" should be "Multiplier"

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